Mapping With Melanin™ — Map Experience UX Specification

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1. Non-Negotiable Rules

These rules are **product-level constraints** that must never be overridden by feature development, performance optimization, or third-party integration decisions. Any proposed change that conflicts with these rules requires explicit founder approval before implementation.

#	Rule	Rationale
NNR-01	Welcoming scores and caution indicators are based exclusively on **community minority experiences** — how welcomed, respected, and safe community members report feeling. They are **never** derived from police crime statistics, arrest data, census income levels, or any data source that criminalizes minority communities.	Using policing data would reproduce systemic bias and harm the very communities the app serves.
NNR-02	Neighborhoods and areas are framed as **“welcoming”** or **“unwelcoming,”** never as “safe” or “unsafe.”	“Safe/unsafe” language carries implicit criminalization. “Welcoming” centers the community’s experience.
NNR-03	Kinfolk AI references **behavior only** in recommendations (“Since you’ve been browsing salons…”). It **never** references a user’s identity, race, gender, or age (“Since you’re Black…”). This is the **One-Way Mirror Rule**.	Identity-based targeting is discriminatory and a legal risk. Behavior-based targeting is both ethical and effective.
NNR-04	Non-minority (allied) businesses are **never** promoted above minority-owned alternatives in any ranking, recommendation, or featured placement.	The app’s core mission is amplifying minority-owned businesses. Allied businesses may be listed but never elevated.
NNR-05	Community entrepreneurs list their businesses **free, always.** No paywall, no freemium tier, no “upgrade to be visible.”	Charging someone with a dream to be found by their own community is antithetical to the mission.
NNR-06	The app serves the **full diaspora** — Black, Hispanic, Ethiopian, Caribbean, Indigenous, Vietnamese, Korean, Brazilian, West African, Haitian, and all other minority communities. It is not a “Black-owned business app.”	Exclusivity within the diaspora replicates the same harm the app was built to counter.
NNR-07	The **Sundown Towns / Historical Caution Zones** layer is **always visible** and cannot be turned off. It is presented with dignity and historical context, never as a fear-mongering alert.	This information protects community members. Hiding it would be a safety failure.
NNR-08	**Active Safety Alerts** are always visible and always the highest z-index element on the map. No UI element may	Active alerts are life-safety information.

#	Rule	Rationale
	obscure them.	
NNR-09	Reporter identity for Sundown Town / Caution Zone reports is **always anonymized**. No report may be traceable to a specific user.	Reporters may face retaliation. Privacy protection is non-negotiable.
NNR-10	The word **“dangerous”** is permanently banned from all user-facing copy. The word **“unsafe”** is banned from all community-facing descriptions of areas or neighborhoods.	Language shapes perception. These words criminalize space rather than describe experience.

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2. Core Insight & Design Philosophy

> *“I just want to know where to shop and where to avoid.”* — MWM Community Member

This single sentence defines the **minimum viable experience**. The map must work perfectly for a user who never explores beyond this need. Every additional feature — cultural sites, Kinfolk recommendations, Living Legacy, community circles — is a layer that reveals itself naturally over time. The user is never forced to engage with depth they haven’t asked for.

The design philosophy rests on three principles:

Clarity before depth. The default state of the map is clean, readable, and immediately useful. A user in a new city should be able to open the app and find a trusted hair salon within thirty seconds without reading a tutorial or configuring anything.

Trust before engagement. The app earns the right to offer more. Kinfolk does not speak until it has observed enough behavior to say something genuinely useful. Layers do not appear until the user has demonstrated curiosity. The community’s data is treated as sacred, not as a product feature.

Dignity in every pixel. Historical trauma, community caution, and cultural heritage are presented with the weight they deserve. Nothing is gamified that shouldn’t be. Nothing is minimized that matters.

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3. Default Map View

3.1 What the User Sees on First Open

When a user opens the map tab for the first time — or any subsequent time without active filters — they see exactly the following, and nothing more:

Element	Specification
Map base	Standard street map, dark mode default (respects system preference). No satellite view by default.
Business pins	Minority-owned businesses within a **5–10 mile radius** of the user's current location only. Default radius is 7 miles; adjustable in settings.
Active Safety Alerts	Any active alerts in the visible map area, always displayed regardless of zoom or radius.
Sundown Town / Caution Zone boundaries	Subtle faded amber boundary lines for any documented zones within the visible area. Always present, never prominent.
Layer hint	A small "Layers" button in the bottom-right corner with a subtle pulsing dot on first open only, indicating that more content is available. The dot disappears after the user taps it once.
Search bar	Persistent at the top. Placeholder text: *"Find businesses, places, or communities…"*
Kinfolk button	Small, bottom-left corner. Inactive state (no badge, no animation) until Kinfolk has something to say.

3.2 What Is Explicitly NOT Shown by Default

The following elements are **never** shown in the default view without user action:

- Cultural sites, murals, or landmarks
- Heritage district boundaries
- HBCU locations
- Farmers markets
- Community events
- Welcoming indicators
- Businesses outside the active radius
- Any Kinfolk recommendation panel

3.3 Radius Loading Behavior

The map loads pins only for the **visible viewport plus a 20% buffer** beyond the screen edges. As the user pans, new pins load in the direction of movement. Pins that scroll more than 2× the screen width away from the viewport are unloaded from memory. This is the primary mechanism preventing the "2,220 pins at once" problem.

The radius filter (5–10 miles) applies to **initial load and search results**, not to manual panning. If a user manually pans to a different city, the map loads that area's pins on demand with a brief loading indicator.

3.4 The "More Content" Indicator

The layer hint button uses a **single pulsing amber dot** (not a notification badge, not a number) to signal depth without demanding attention. The animation runs for 3 seconds on first open, then stops. It reappears only when a new layer type becomes available that the user has never seen (e.g., when a community event is added near them for the first time).

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4. Always-On Layers

These two layers are **hardcoded on**. They cannot be toggled off in the layers panel. The layers panel UI must not include a toggle for either of them — their presence is acknowledged with a static label: *"Always active — community safety."*

4.1 Active Safety Alerts / Emergency Alerts

Property	Specification
Trigger	Community-submitted or admin-verified active safety event (e.g., reported incident of hostility, active protest, road closure affecting community safety)
Visual	Large red circle with `!` icon, pulsing ring animation, always highest z-index
Tap behavior	Full-screen bottom sheet: alert title, description, time posted, number of community confirmations, option to confirm or dismiss
Expiry	Alerts auto-expire after 24 hours unless renewed by admin or re-confirmed by ≥5 community members
Push notification	If user is within 2 miles of a new active alert, push notification is sent (respects notification preferences)

4.2 Sundown Towns / Historical Caution Zones

See **Section 5** for the full detailed specification. The summary for this section:

- Always visible as a subtle faded amber boundary or small archive icon
- Never alarming, never prominent, never the first thing the eye goes to
- Tapping opens a full context card with historical record and recent community reports
- The visual weight of the indicator changes based on the zone's current state (see Section 5)

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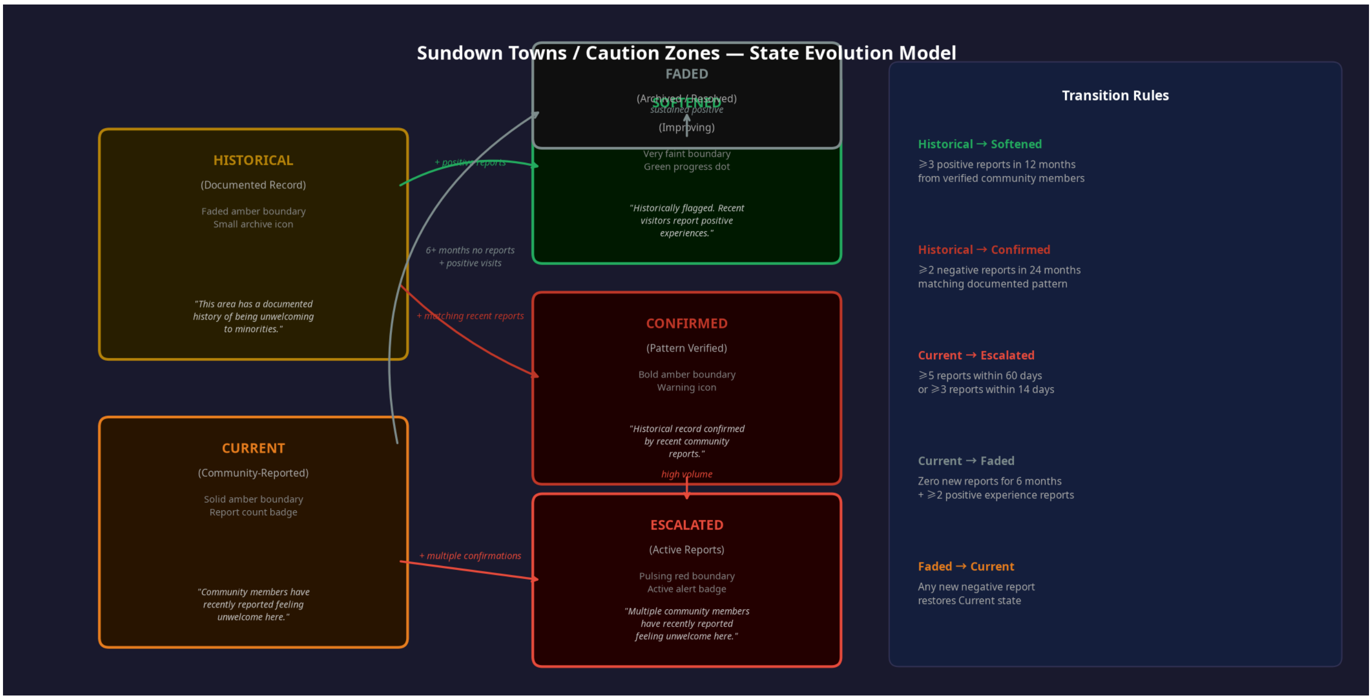
5. Sundown Towns — Detailed Specification

5.1 Overview

Sundown Towns are areas with documented or community-reported histories of being unwelcoming or hostile to minority communities. The MWM app treats this data with the gravity it deserves: it is presented as historical and community context, not as a fear alert. The goal is to **inform with dignity**, not to frighten.

The system distinguishes between two fundamentally different types of caution zones, each with its own visual treatment, language, and evolution rules.



5.2 Zone Types

Type A: Historical Sundown Town (Documented Record)

A Historical zone is derived from documented historical records — academic research, civil rights archives, local historical societies, or the Sundown Towns database maintained by researchers such as James Loewen's work. These are facts of record, not community opinion.

Visual Treatment:

Property	Specification
Boundary line	Faded amber dashed line, 40% opacity
Fill	Very light amber wash, 8% opacity
Icon	Small archive/book icon at zone centroid, 16px, amber at 50% opacity
Label	No text label on map — label appears only in tap card

Default Sample Language (tap card header):

> *"This area has a documented history of being unwelcoming to minorities. Community members can share their current experience to help update this indicator."*

Sub-states of Historical:

Sub-state	Trigger	Visual Change	Sample Language
Historical — Neutral	Default state; no recent reports	Faded amber boundary (base state)	*(as above)*
Historical — Softened	≥3 positive community reports within 12 months	Boundary fades to 20% opacity; green progress dot at centroid	*"This area has a documented history of being unwelcoming to minorities. Recent visitors report positive experiences. Community members continue to share updates."*
Historical — Confirmed	≥2 negative community reports within 24 months matching documented pattern	Boundary increases to 60% opacity; amber warning dot at centroid	*"This area has a documented history of being unwelcoming to minorities. Recent community members have reported experiences consistent with that history."*

Type B: Current Community-Reported Area (Present-Day Reports)

A Current zone is created entirely from present-day community member reports. No historical record is required. Any community member can submit a report; the system aggregates reports to determine zone state.

Visual Treatment:

Property	Specification
Boundary line	Solid amber line, 70% opacity
Fill	Amber wash, 15% opacity
Icon	Small community/people icon at zone centroid, 18px, amber at 80% opacity
Report count badge	Small number badge on the icon showing report count (e.g., “4 reports”)
Label	No text label on map

Default Sample Language (tap card header):

> *“Community members have recently reported feeling unwelcome here. [X] reports in the last [timeframe].”*

Sub-states of Current:

Sub-state	Trigger	Visual Change	Sample Language
Current — Active	Default state; 1–4 reports, no escalation threshold met	Solid amber boundary (base state)	*”[X] community members have recently reported feeling unwelcome in this area.”*
Current — Escalated	≥5 reports within 60 days, OR ≥3 reports within 14 days	Boundary becomes bold amber, 90% opacity; pulsing dot at centroid; push notification to users within 3 miles	*“Multiple community members have recently reported feeling unwelcome in this area. [X] reports in the last [timeframe]. Please be aware and trust your instincts.”*
Current — Faded	Zero new reports for 6+ months AND ≥2 positive experience reports	Boundary fades to 15% opacity; grey archive icon replaces amber icon; report count hidden	*“No recent reports for this area. [X] community members previously reported concerns. This indicator will be fully archived if no new reports are received.”*

5.3 Zone Evolution: State Transition Rules

The following table defines every valid state transition and the exact conditions that trigger it. All transitions are computed server-side on a nightly batch job, with real-time escalation checks running every 15 minutes for the Escalated trigger.

From State	To State	Condition	Who Can Trigger
Historical — Neutral	Historical — Softened	≥3 verified positive reports in 12 months	Verified community members
Historical — Neutral	Historical — Confirmed	≥2 verified negative reports in 24 months	Verified community members
Historical — Softened	Historical — Neutral	Positive reports drop below threshold (reports age out after 18 months)	Automatic (system)
Historical — Confirmed	Current — Escalated	≥5 total reports (historical + current) within 60 days	Verified community members
Current — Active	Current — Escalated	≥5 reports in 60 days OR ≥3 reports in 14 days	Verified community members
Current — Active	Current — Faded	0 new reports for 6 months + ≥2 positive reports	Automatic (system) + community members
Current — Escalated	Current — Active	Report rate drops below escalation threshold for 30 days	Automatic (system)
Current — Faded	Current — Active	Any new negative report submitted	Verified community members
Current — Faded	Fully Archived	0 new reports for 18 months + ≥5 positive reports	Automatic (system)

> **Note:** A “verified community member” is any user who has completed phone verification and has been active in the app for at least 7 days. This prevents coordinated false reporting.

5.4 The Tap Card (Full Context View)

When a user taps any caution zone boundary or icon, a bottom sheet slides up with the following structure:

Section 1 — Zone Header

- Zone type badge: “Historical Record” (amber, archive icon) or “Community Reports” (amber, people icon)
- Zone name (if named) or general description (“Area near [landmark/intersection]”)

- Current state label (e.g., “Softened,” “Escalated”)
- Last updated timestamp

Section 2 — Context Statement

- The appropriate sample language for the current state (see Section 5.2)
- For Historical zones: a brief 2–3 sentence historical summary drawn from the source record
- Source attribution: “Source: [Research institution / Historical archive]” — never “Source: [Username]”

Section 3 — Community Reports Summary

- Number of reports in the last 30 / 90 / 365 days (three separate counts)
- Breakdown: positive experiences vs. negative experiences (no individual reports shown)
- A bar showing the ratio of positive to negative reports over time (simple horizontal bar)

Section 4 — Share Your Experience (CTA)

- Button: “Share my experience here”
- Opens a report form (see Section 5.5)

Section 5 — What This Means

- Collapsible section explaining how MWM caution zones work
- Includes the statement: *“MWM caution zones are based entirely on community experiences — how welcomed and respected our members report feeling. They are never based on crime statistics or police data.”*

5.5 Reporting Form

Field	Type	Notes
Experience type	Toggle: “I felt welcome” / “I felt unwelcome”	Required
When	Date picker: “Today,” “This week,” “This month,” “Earlier”	Required
What happened (optional)	Free text, 280 char max	Optional; never shown publicly
Would you return?	Toggle: Yes / Unsure / No	Optional

Privacy rules:

- No report is ever attributed to a username, profile, or any identifying information in any public-facing view.
- Report text is stored encrypted and visible only to MWM admins for moderation purposes.
- Aggregate counts (e.g., “4 reports”) are the only public-facing data.
- Users may delete their own reports at any time from their profile settings.

5.6 Permanently Banned Language

The following words and phrases are **banned from all user-facing copy** related to caution zones, areas, or neighborhoods:

Banned	Use Instead
“dangerous”	“community members have reported feeling unwelcome”
“unsafe”	“unwelcoming”
“high crime”	*(never reference crime at all)*
“bad area”	“historically flagged area”
“avoid” (as a directive)	“be aware” / “trust your instincts”
“risky”	“community members have shared concerns”

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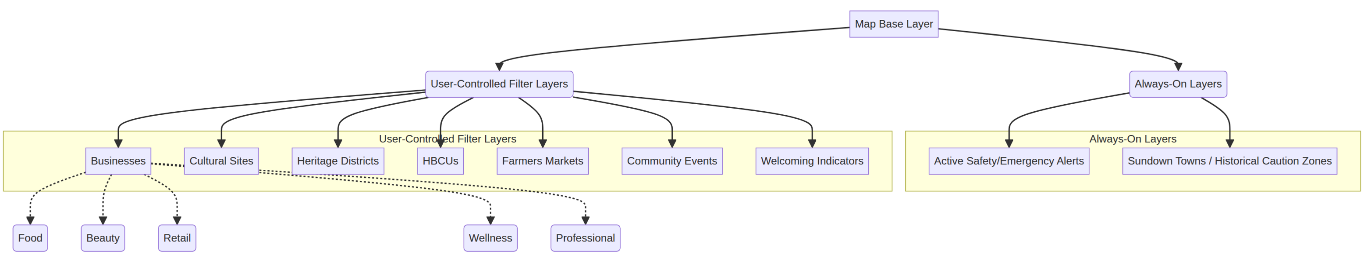
6. User-Controlled Filter Layers

These layers are accessed via the **Layers panel** (bottom-right button). Each layer has an on/off toggle. All layers are **off by default** except Businesses, which is on by default as it is the core use case.



6.1 Layer Definitions

Layer	Default State	Sub-filters	Pin/Visual Style	Notes
Businesses	ON	Food & Dining, Beauty & Wellness, Retail, Professional Services, Health, Entertainment, Other	Medium green circle, category icon	Primary layer. Always loads first.
Cultural Sites	OFF	Museums, Murals & Public Art, Landmarks, Historic Sites	Medium purple star	Loads on demand
Heritage Districts	OFF	—	Soft purple boundary fill with label	Polygon overlay, not a pin
HBCUs	OFF	—	Medium blue graduation cap icon	Point pin
Farmers Markets	OFF	—	Small-medium green leaf icon	Includes operating hours in pin card
Community Events	OFF	This Week, This Month	Medium red diamond, time badge	Time-bound; auto-removes when event ends
Welcoming Indicators	OFF	—	Small teal checkmark	Overlaid on business pins that have earned the indicator

6.2 Layer Interaction Rules

When multiple layers are active simultaneously, the following rules govern display priority:

1. Active Safety Alerts always render at the highest z-index, above all other layers.
2. Sundown Town / Caution Zone boundaries render below all pin layers but above the base map.
3. Heritage District polygon fills render below all pins.
4. Business pins render above cultural and event pins when they overlap.

5. Welcoming indicators render as overlays on top of the business pin they belong to, not as separate pins.

6.3 Filter Persistence

The user’s active layer state is saved to their profile and persists across sessions and devices. The first time a user activates a layer, a brief one-line tooltip appears below the layer button: *”[Layer name] added to your map. Tap any pin to learn more.”* This tooltip auto-dismisses after 4 seconds and never appears again for that layer.

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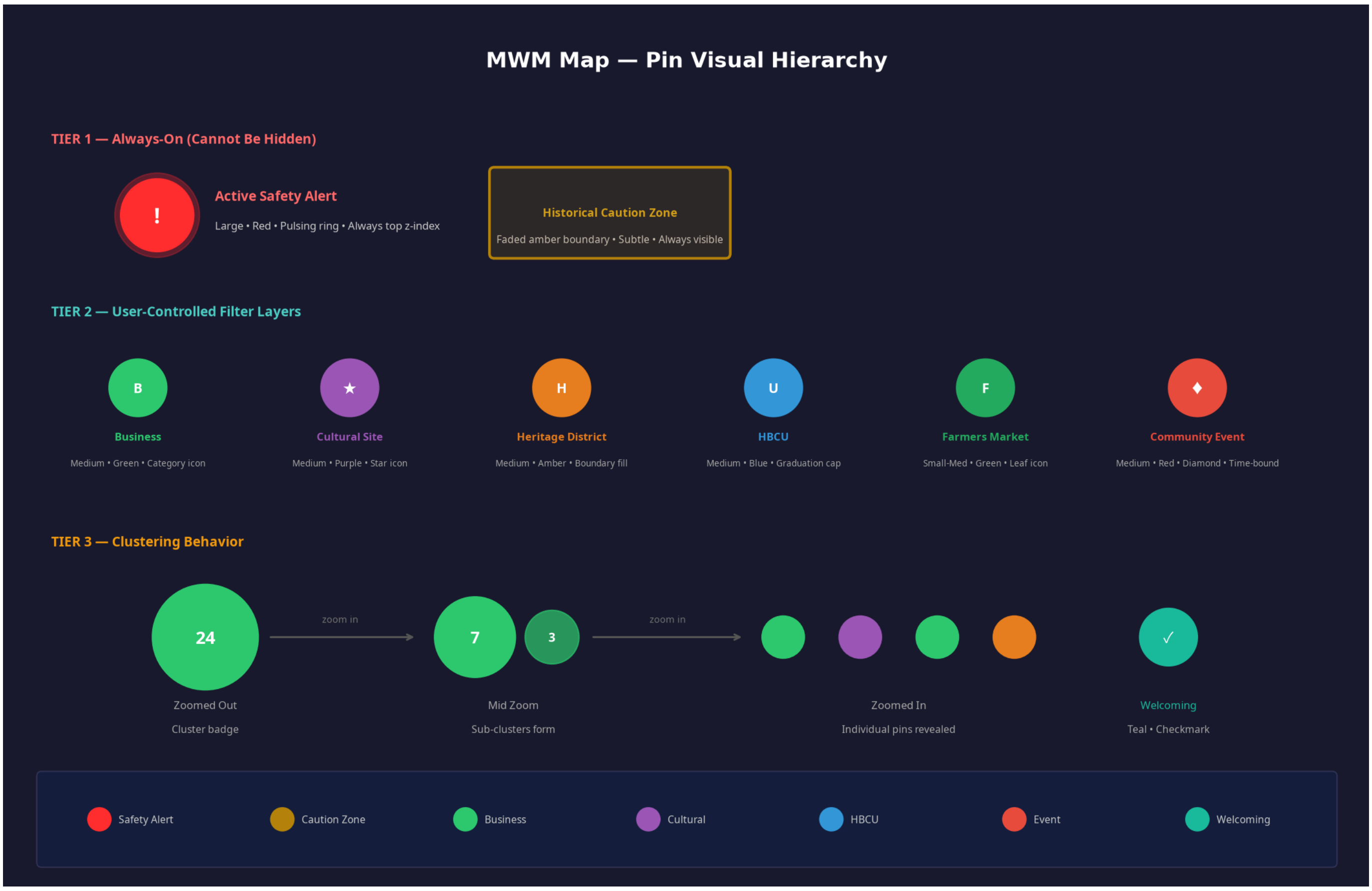
7. Pin Visual Hierarchy



7.1 Pin Specifications by Type

Pin Type	Shape	Size (mobile)	Primary Color	Icon	Z-Index	Animation
Active Safety Alert	Circle	44px	`#FF2D2D` Red	`!` bold	100	Pulsing ring
Selected Pin (any type)	Circle	52px (enlarged)	Inherits type color	Inherits type icon	90	Scale-up on tap
Business — Food	Circle	32px	`#2ECC71` Green	Fork/knife	50	None
Business — Beauty	Circle	32px	`#2ECC71` Green	Scissors	50	None
Business — Retail	Circle	32px	`#2ECC71` Green	Shopping bag	50	None
Business — Wellness	Circle	32px	`#2ECC71` Green	Leaf	50	None
Business — Professional	Circle	32px	`#2ECC71` Green	Briefcase	50	None
Cultural Site	Circle	30px	`#9B59B6` Purple	Star	40	None
HBCU	Circle	30px	`#3498DB` Blue	Graduation cap	40	None
Farmers Market	Circle	28px	`#27AE60` Dark Green	Leaf	35	None
Community Event	Diamond	30px	`#E74C3C` Red	Diamond	45	Subtle pulse while active
Welcoming Indicator	Overlay badge	14px	`#1ABC9C` Teal	Checkmark	55	None
Historical Caution — Neutral	Boundary + icon	16px icon	`#B8860B` Amber	Archive	10	None

Pin Type	Shape	Size (mobile)	Primary Color	Icon	Z-Index	Animation
Historical Caution — Confirmed	Boundary + icon	18px icon	`#B8860B` Amber	Warning	15	None
Current Caution — Active	Boundary + icon	18px icon	`#E67E22` Amber	People	20	None
Current Caution — Escalated	Boundary + icon	20px icon	`#E74C3C` Red-Amber	People	25	Slow pulse

7.2 Clustering Behavior

Clustering prevents visual overload when many pins are close together. The following rules govern clustering at each zoom level:

Zoom Level	Behavior	Cluster Appearance
City view (zoom 10–11)	All business pins cluster into single badges per neighborhood	Large circle with count (e.g., “24”), colored by dominant pin type
District view (zoom 12–13)	Clusters break into sub-neighborhood groups	Medium circle with count (e.g., “7”)
Street view (zoom 14–15)	Clusters of 3+ pins within 40px of each other merge	Small circle with count (e.g., “3”)
Building view (zoom 16+)	All individual pins shown	No clustering

Cluster tap behavior: Tapping a cluster badge zooms the map to the next level down, centering on that cluster. It does not open a list view. If the cluster contains only 2–3 pins at maximum zoom, tapping opens a mini-list card showing all pins in the cluster.

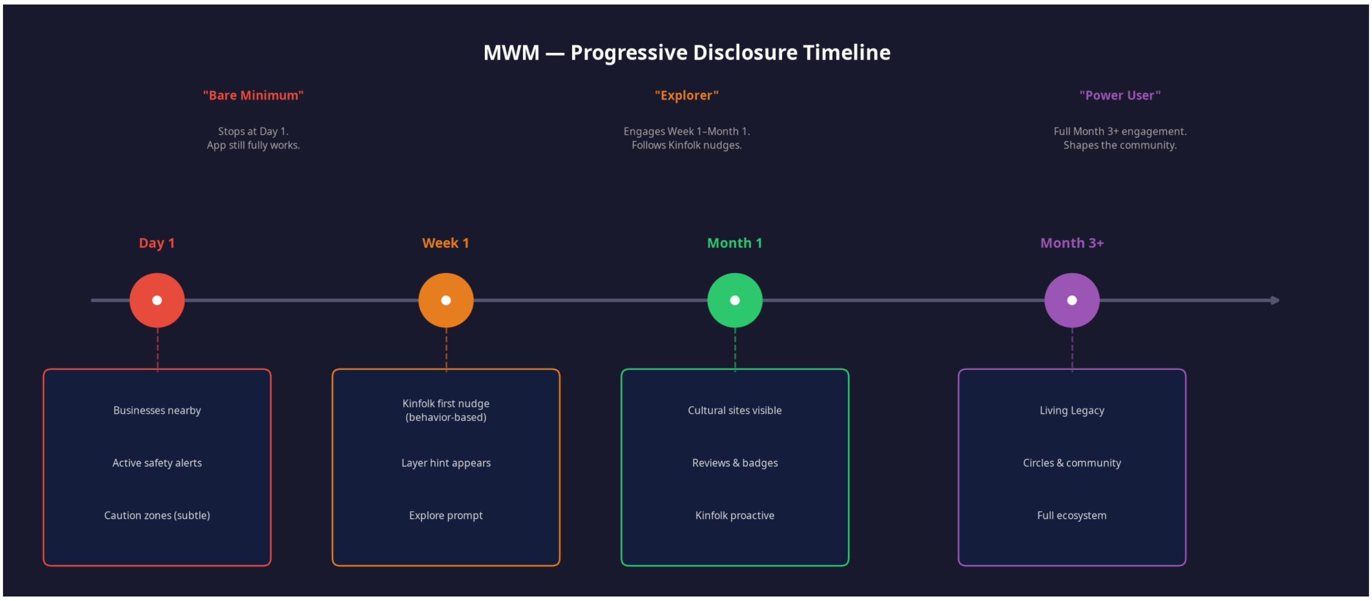
Safety Alert clustering exception: Active Safety Alert pins **never** cluster. They always display as individual pins regardless of zoom level or proximity to other pins.

7.3 Preventing Visual Clutter

When two or more pins of different types overlap at the same location (e.g., a business that is also a cultural landmark), the following rules apply:

1. Display the highest-priority pin type (per z-index table above).
2. Add a small stacked-layers indicator (two overlapping circles, 8px) in the bottom-right of the pin.
3. Tapping the pin opens a card that lists all entities at that location with individual tap targets.

8. Progressive Disclosure



The app never overwhelms. Depth reveals itself in response to demonstrated interest and elapsed time. The following schedule governs how features surface:

8.1 Disclosure Timeline

Phase	Trigger	What Appears	What Does Not Appear
Day 1	First open	Businesses (5–7mi radius), Active Safety Alerts, Caution Zones (subtle), Layer hint dot	All other layers, Kinfolk panel, reviews, badges
Day 3–7	User has tapped ≥3 business pins	Layer hint dot pulses once more; Kinfolk shows a single passive nudge in the bottom bar (see Section 11)	Kinfolk panel, full recommendations
Week 2	User has used search or tapped a cultural site	Cultural Sites layer becomes highlighted in the Layers panel with a “New” badge	Heritage districts, HBCUs, events
Month 1	User has left ≥1 review OR visited ≥10 businesses	Review prompts appear on business cards; badge system introduced via a single in-app notification	Living Legacy, Circles
Month 3+	User has ≥5 reviews OR joined a Circle	Full ecosystem: Living Legacy contributions, Circle map overlays, Kinfolk proactive briefings	Nothing — all features now available

8.2 The Aha Moment

The designed “aha moment” — the instant a user realizes the app has depth — occurs during **Day 3–7**, when Kinfolk surfaces its first nudge. The nudge is a single line in the bottom bar, not a modal, not a notification:

> *“Since you’ve been browsing salons, there’s a highly-rated one 0.4 miles from here.”*

This is the first time the user realizes the app has been paying attention. It is not intrusive. It is not demanding. It simply demonstrates that the app knows them — through their behavior, not their identity.

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9. Three User Personas

These personas define the **full range of valid user journeys**. The app must work perfectly for all three without requiring any persona to change their behavior.

9.1 The Bare Minimum User

Profile: Opens the app, finds businesses, checks caution zones, leaves. Never interacts with Kinfolk, never leaves a review, never explores layers. May use the app daily for years in this mode.

What the app provides:

- Clean, fast map with nearby businesses
- Caution zone awareness without alarm
- Active safety alerts when relevant
- Search that works immediately

What the app never does to this user:

- Prompts them to leave a review more than once per business visit
- Shows Kinfolk recommendations unprompted after the first nudge is dismissed
- Adds visual complexity to their map
- Sends push notifications about features they haven't engaged with

Success metric: This user finds a trusted business within 60 seconds of opening the app, every time.

9.2 The Explorer

Profile: Curious. Taps cultural sites when they appear. Reads the city stories. Follows Kinfolk's recommendations occasionally. May leave a review when the experience was exceptional. Engages with the app 3–5 times per week.

What the app provides:

- Gentle layer introductions based on behavior
- Kinfolk nudges that feel like a knowledgeable friend, not an algorithm
- Cultural context that enriches their understanding of the places they visit
- A sense that the community is alive and contributing

What the app never does to this user:

- Overwhelms them with all features at once
- Requires them to configure anything to get value
- Makes them feel behind or missing out on features

Success metric: This user discovers at least one new business or cultural site per week that they would not have found otherwise.

9.3 The Power User

Profile: Leaves reviews regularly. Has earned badges. Participates in at least one Circle. Has contributed to Living Legacy. Engages with Kinfolk daily. Treats the app as a community platform, not just a map.

What the app provides:

- Proactive Kinfolk briefings (opt-in)
- Route planning with cultural and business stops
- Circle map overlays showing where their community has been
- Living Legacy contribution tools
- Safety briefings when traveling to new cities

What the app never does to this user:

- Reduces their influence or visibility because they haven't paid
- Shares their data with businesses in an identifiable way
- Promotes non-minority businesses above their recommendations

Success metric: This user contributes at least one piece of community data (review, report, Living Legacy entry) per week and feels that their contribution shapes the experience for others.

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10. First-Time Experience

10.1 The Map on First Open

The sequence of events when a new user opens the map tab for the first time:

1. **Location permission prompt** (if not already granted): Standard iOS/Android permission dialog. No custom pre-prompt. If denied, the map defaults to the user's city based on IP geolocation with a banner: *"Enable location for a more accurate experience."*
2. **Map loads** with a brief skeleton state (0.3–0.8 seconds): Grey placeholder circles where business pins will appear, in the correct geographic positions. This prevents the jarring "empty map then sudden pins" experience.
3. **Pins fade in** over 0.4 seconds as data loads. Pins load nearest-first (closest to user location appears first).
4. **Layer hint dot** appears on the Layers button with a single 3-second pulse animation.
5. **No tutorial overlay, no onboarding modal, no walkthrough.** The map is immediately usable.

10.2 Teaching Without a Tutorial

The app teaches through **contextual discovery**, not instruction:

Discovery Moment	How It Teaches
User taps a business pin	Bottom sheet slides up showing business details. First tap includes a subtle “Tap any pin to learn more” hint text below the sheet, auto-dismissed after 4 seconds. Never shown again.
User taps the Layers button	Layers panel slides up. Each layer has a one-line description. No tutorial needed — the labels are self-explanatory.
User taps a caution zone	Context card slides up with historical information. The card itself explains what the zone means.
User taps Kinfolk button	If Kinfolk has nothing to say yet, a brief message appears: *“I’m learning your preferences. Check back after you’ve explored a bit.”*
User searches for something	Results appear on map as highlighted pins. A “Clear search” button appears. No instruction needed.

10.3 The Aha Moment (Detailed)

The aha moment is engineered to occur naturally, not forced. It happens when Kinfolk surfaces its first behavior-based nudge (see Section 8.2). The design requirements for this moment:

- The nudge appears as a **single line of text in the bottom navigation bar**, not a modal or notification
- It is dismissible with a single swipe down
- If dismissed, it does not reappear for 48 hours
- If tapped, it opens the recommended business pin directly on the map
- The language is warm and specific, never generic: *“Since you’ve been browsing salons, there’s a highly-rated one 0.4 miles from here”* — not *“We found businesses you might like”*

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11. Kinfolk’s Role in the Map

Kinfolk is the app’s AI community guide. On the map, Kinfolk’s role is defined by a strict principle: **it earns the right to speak**. It does not speak until it has something genuinely useful to say, and it never speaks in a way that feels like surveillance.

11.1 Kinfolk Behavior by Persona

Persona	Kinfolk Behavior	Trigger	Sample Language
Bare Minimum	Silent by default. Responds only when directly asked.	User taps Kinfolk button	*“What are you looking for? I can help you find businesses, check an area, or plan a route.”*
Explorer	Gentle, single-line nudges in the bottom bar. Maximum 1 per session.	≥3 taps on same category of business	*“Since you’ve been browsing salons, there’s a highly-rated one 0.4 miles from here.”*
Power User	Proactive briefings (opt-in). Route planning. Pre-trip safety context.	User opts in to proactive mode OR opens map in a new city	*“You’re in Atlanta. Here’s what the community is saying about this area this week.”*

11.2 The One-Way Mirror Rule (Enforcement)

Every Kinfolk message must pass the following test before it is sent:

> **Does this message reference the user’s identity (race, gender, age, religion, national origin)?** If yes: **reject and rewrite.** If no: **proceed.**

Valid: *“Since you’ve been browsing Ethiopian restaurants…”* Invalid: *“As an Ethiopian community member…”*

Valid: *“Since you visited this salon last week…”* Invalid: *“Since you’re a Black woman looking for natural hair care…”*

This rule applies to all Kinfolk outputs: map nudges, route suggestions, safety briefings, and any AI-generated text.

11.3 What Kinfolk Never Does on the Map

- Opens a full panel unprompted (bare minimum and explorer users)
- Sends a push notification about a recommendation (only safety alerts trigger push)
- Promotes a non-minority business above a minority-owned alternative
- References a user’s demographic data
- Speaks more than once per session to a user who has not engaged with its previous message

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12. Technical Requirements for Replit

12.1 Map Performance: Handling 2,220+ Pins

The core performance strategy is **never loading all pins at once**. The following architecture must be implemented:

Strategy	Implementation
Viewport-based loading	Query the backend for pins within `viewport_bounds + 20% buffer` only. Re-query on pan/zoom end (debounced 300ms).
Radius filter on initial load	On app open, load only pins within the user's configured radius (default 7 miles).
Tile-based caching	Divide the map into geographic tiles (e.g., 0.1° × 0.1° grid). Cache tile data in memory for 5 minutes. Re-use cached tiles on pan-back.
Layer-gated queries	Only query the backend for a layer's pins when that layer is toggled on. Do not pre-fetch inactive layers.
Unload distant pins	Pins more than 2× the screen width from the current viewport center are removed from the render tree (not from cache).

12.2 Clustering Algorithm

Use **supercluster** (the standard library used by Mapbox and react-map-gl) for client-side clustering. Configuration:

```
const cluster = new Supercluster({
  radius: 60,         // pixel radius for clustering
  maxZoom: 15,        // stop clustering at zoom 16+
  minZoom: 0,
  minPoints: 3,       // minimum pins to form a cluster
});
```

Safety Alert exception: Filter Safety Alert pins out of the supercluster input array and render them separately, always unclustered, always at the highest z-index.

12.3 Radius-Based Loading

```
// Pseudocode for radius-based pin loading
async function loadPinsForViewport(viewport, userLocation, activeFilters) {
  const bounds = expandBounds(viewport, 0.20); // 20% buffer
  const radiusFilter = userSettings.radius || 7; // miles

  const query = {
    bounds,
    radius_miles: radiusFilter,
    center: userLocation,
    layers: activeFilters, // only active layers
    limit: 500,            // max pins per request
    offset: paginationCursor,
  };

  const pins = await api.getPins(query);
  renderPins(pins);
}
```

The backend must support a `bounds` + `radius` + `layers` compound query with an index on `(latitude, longitude, layer_type, is_active)`.

12.4 Filter State Persistence

User layer preferences are stored in two places:

Storage	Purpose	TTL
`AsyncStorage` (local)	Immediate persistence; survives app restart without network	Indefinite
User profile (server)	Cross-device sync; loaded on login	Indefinite

On app open: load from `AsyncStorage` immediately (no network wait), then sync with server in background. If server state differs from local state, server state wins and local state is updated silently.

12.5 Offline Behavior

Scenario	Behavior
Full offline (no connection)	Show last-cached map tiles and pins. Display banner: *"You're offline. Showing your last saved map."*
Partial connection (slow/intermittent)	Show cached data; attempt background refresh; show subtle spinner on map edge
Safety Alerts offline	Last-known active alerts remain visible with a timestamp: *"Last updated [time]. Connect to get current alerts."*
Sundown Town data offline	Always available — this data is bundled with the app and updated on install/update, not fetched live
Search offline	Search is disabled; show: *"Search requires a connection. Browsing your saved map."*
Pin tap offline	Show cached business card data if available; show *"Full details unavailable offline"* if not

Offline data bundling: The Sundown Towns / Historical Caution Zones dataset must be bundled with the app binary and updated via app update, not via live API call. This ensures caution zone data is always available regardless of connection state.

12.6 Database Index Requirements

The following indexes are required for acceptable map query performance:

```
-- Primary spatial index for pin queries
CREATE INDEX idx_pins_location ON pins USING GIST (location);

-- Compound index for layer + active queries
CREATE INDEX idx_pins_layer_active ON pins (layer_type, is_active, created_at);

-- Index for radius queries
CREATE INDEX idx_pins_lat_lng ON pins (latitude, longitude);

-- Index for caution zones (polygon queries)
CREATE INDEX idx_caution_zones_boundary ON caution_zones USING GIST (boundary_p
```

12.7 API Endpoint Specifications

Endpoint	Method	Parameters	Response
`/api/map/pins`	GET	`bounds`, `radius_miles`, `center_lat`, `center_lng`, `layers[]`, `limit`, `offset`	Array of pin objects with `id`, `lat`, `lng`, `type`, `layer`, `name`, `is_active`
`/api/map/caution-zones`	GET	`bounds`	Array of zone objects with `id`, `boundary_geojson`, `zone_type`, `state`, `report_count`
`/api/map/safety-alerts`	GET	`bounds`	Array of alert objects with `id`, `lat`, `lng`, `title`, `description`, `expires_at`, `confirmation_count`
`/api/map/zones/:id/reports`	GET	`zone_id`	Aggregated report counts by period and sentiment
`/api/map/zones/:id/report`	POST	`zone_id`, `experience_type`, `when_period`, `would_return`	Confirmation

---

13. Appendix: Sample Copy Strings

This appendix provides the canonical copy for every user-facing string related to caution zones, Kinfolk, and layer interactions. Development must use these exact strings. Any variation requires founder approval.

13.1 Caution Zone States — Full Copy

State	Tap Card Header	Tap Card Body	Map Tooltip (on hover/long-press)
Historical — Neutral	“Documented History”	“This area has a documented history of being unwelcoming to minorities. Community members can share their current experience to help update this indicator.”	“Historical caution zone — tap to learn more”
Historical — Softened	“Improving Area”	“This area has a documented history of being unwelcoming to minorities. Recent visitors report positive experiences. Community members continue to share updates.”	“Community members report positive recent experiences — tap for history”
Historical — Confirmed	“Pattern Confirmed”	“This area has a documented history of being unwelcoming to minorities. Recent community members have reported experiences consistent with that history.”	“Historical pattern confirmed by recent reports — tap to learn more”
Current — Active	“Community Reports”	“Community members have recently reported feeling unwelcome in this area. [X] reports in the last [timeframe].”	”[X] community reports — tap to learn more”
Current — Escalated	“Active Community Concern”	“Multiple community members have recently reported feeling unwelcome in this area. [X] reports in the last [timeframe]. Please be aware and trust your instincts.”	“Active community concern — [X] recent reports”
Current — Faded	“Archived Reports”	“No recent reports for this area. Community members previously shared concerns here. This indicator will be fully archived if no new reports are received.”	“Archived — no recent reports”

13.2 Kinfolk Map Nudges — Sample Library

Trigger Behavior	Sample Nudge
Browsed ≥3 salon pins	"Since you've been browsing salons, there's a highly-rated one 0.4 miles from here."
Browsed ≥3 food pins	"Since you've been exploring restaurants, a community favorite is open right now nearby."
Tapped a cultural site	"There are two more murals within walking distance. Want to see them?"
Opened map in a new city	"You're in [City]. Here's what the community is saying about this area this week."
Searched for a business type	"I found [X] [business type] results near you. The top-rated one is [distance] away."
User has been inactive 30+ days	*(No nudge. Kinfolk does not nag.)*

13.3 Offline State Messages

Scenario	Message
Full offline	"You're offline. Showing your last saved map."
Safety alert stale	"Last updated [time ago]. Connect to get current alerts."
Search disabled	"Search requires a connection. Browsing your saved map."
Business details unavailable	"Full details unavailable offline. Connect to see reviews and hours."

End of Specification

Document maintained by: Mapping With Melanin™ Product Team
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Change log: v1.0 — Initial release. v1.1 — Added Sundown Towns historical/current distinction, full state transition rules, and expanded sample copy library per founder direction (August 7, 2026).